SAMPLE

Earth's Sky: The Pole Stars

Polaris, Navigation, and the Celestial Poles

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Introduction

Of all the subjects in this series, the one this volume treats is the sky's one fixed point, and the book begins with the paradox that the fixity is approximate and the point is plural. A pole star is the star that stands nearest the celestial pole, the hub around which the whole sky appears to turn, the one point of the northern or southern heavens that a long exposure photographs as a dot while every other star draws an arc; and because the Earth's axis both wobbles and drifts against the stars, the title has been passed from star to star across the millennia and will be passed again, the office being permanent and its holders being temporary. The current incumbent of the north is Polaris, a bright yellow supergiant at the end of the Little Dipper's handle, celebrated in navigation, hymnody, and flag design; the south's holder is the faint Sigma Octantis, a star so dim that most of the hemisphere it serves has never seen it, the two poles being a matched pair of unequal honors.

The series' standing method applies plainly here: the subject treated as a character with a biography, its mechanics explained, its history told, its practical uses set out for the reader who wants them, its unsolved questions kept plainly on the table. The subject's special advantage is that everyone owns a share of it: the pole star is the one star the northern hemisphere's nonspecialists can generally name, and the sky's rotation around it is the one piece of astronomy visible to anyone who stands outside long enough. The volume's quiet argument is that the pole star is the reader's oldest astronomical instrument, older than the telescope and the almanac, the sky's own benchmark for direction, latitude, and time, and that its story, from the pyramids' alignment to the interferometers now weighing Polaris, is the history of practical astronomy in miniature.

The hurried reader's preview: the axis wobbles through a cycle of roughly twenty-six thousand years, discovered by the Greek astronomer Hipparchus in the second century before the common era; the north's succession has run from Thuban, the fourth-magnitude star the pyramid builders knew, through the pole-less gap the Romans endured, to Polaris, closest to the pole around the year 2100, with Vega's turn due in about twelve thousand years; the south, which has never had a bright holder, leans on the Southern Cross and two dim signposts; Polaris itself is a Cepheid variable in a triple system about four hundred light-years away, growing brighter and calmer in ways its observers still dispute; and the reader who closes this book will know how to find the pole, use it, and hand it on.

One note on the volume's scope belongs at the threshold. The pole stars of the other planets, defined the same way by their own axes, appear here only as perspective, the office being a universal institution with local incumbents; the Earth's magnetic poles, which shift by their own laws and serve the compass, are not the celestial poles and are kept distinct, the navigator's true north and magnetic north being different addresses that the almanacs reconcile; and the pole of the sky's coordinate system, the point the catalogs organize around, is the same pivot this book follows, so the reader who finishes the volume will also have learned how the star charts are built, the office being the one astronomical subject whose practical use and mathematical definition coincide.

Chapter 1: The Discovery of The Pole Stars

The discovery of the pole stars runs deeper than the stars themselves, because the first thing discovered was the wheel. The sky rotates: the constellations rise in the east, cross the meridian, and set in the west, night after night, and the rotation has a center, a point in each hemisphere around which the whole procession swings. The earliest observers knew this by looking, the star groups wheeling around a dark or faintly starred pivot, and the knowledge is old enough to be built into stones: passages and alignments in ancient structures point to the celestial pole's neighborhood, and the Fourth Dynasty Egyptians, whose skies had no bright star at the pivot, aligned their monuments with the pole's slow circle in mind, using the pole's unrising circumpolar stars, which never dipped below the horizon, as the sky's fixed furniture.

The bright holder of the title was then, briefly, Thuban. Around the year three thousand before the common era, the slow wobble of the Earth's axis carried the north celestial pole to within a tenth of a degree of that star, the fourth-magnitude pivot of the constellation Draco, the dragon; the pyramid age's great structures have shafts and alignments consistent with a sky organized around it, and the era's astronomers, whoever they were, are the first people known to have had a proper pole star, a single visible star standing within the polar circle's smallest ring. The office passed as it always does: Thuban crept away from the pivot by a hand's width per century, and by the classical era the pole hung in a dim stretch between the two bears, marked by no star brighter than the fifth magnitude.

The Roman era's sky kept the title vacant. Near the turn of the common era, the celestial pole rode nearly midway

between Polaris, then a middling star of the Little Bear well off the pivot, and Kochab, the brighter of the Bear's two guard stars; the navigators of the classical Mediterranean used the whole Little Bear as a pointer to the empty pivot, the constellation serving collectively where no member served alone. The discovery of the office's next holder was therefore gradual, a matter of centuries: Polaris closed on the pole at the wobble's stately pace, entered the polar circle in the early middle ages, and became the unambiguous North Star only as the second millennium wore on, the brightening pivot of the late medieval and early modern sky.

The formal discovery of the system behind the succession belongs to Hipparchus. Comparing his own positional observations with the star records inherited from the earlier Greek astronomers, he found the equinox creeping westward against the stars, the signature of the axis's wobble, and announced the precession of the equinoxes to the ancient world: the discovery that made the pole star a temporary office rather than a permanent tenant, the first mathematical proof that the sky's one fixed point moves. The later chapters of this discovery belong to the mechanicians, Newton's torque calculations and the eighteenth century's refinement of the wobble's period, and to the archaeoastronomers, who traced the office backward through Thuban and forward through the roster of holders to come.

The southern pole's discovery was the discovery of an absence. The hemisphere's first systematic observers, the European navigators of the sixteenth century, found the south celestial pole in the dim octant of sky near the faint constellation now called Octans, with no star of navigational quality anywhere near it, the pivot of the southern wheel being marked by an emptiness that demanded substitute methods; the Southern Cross and the two bright stars Achernar and Canopus became the pole's stand-ins, geometry standing where a star would not, and the southern sky's practical literature ever since has been a literature of pointers, diagonals, and triangles aimed at a blank pivot. The discovery was completed by the cataloguers, who eventually crowned the fifth-magnitude Sigma Octantis as the office's holder, a technicality rather than a landmark.

The chapter closes on the discovery the reader can make, because the pole's circle is drawn in exposure. Point a camera at the northern pivot on a clear night, hold the shutter open for minutes, and the film shows the wheel: every star a concentric arc, the hub a small ring with Polaris riding inside it, the whole rotation of the heavens reduced to a single photograph, the discovery being the one the earliest skywatchers made, available still to anyone with patience, the fixed point of the sky demonstrating itself for free.

The succession's discovery reached its deepest register in the eighteenth and nineteenth centuries, when the wobble became a computed quantity rather than a recorded one. Newton had given the mechanism in his Principia, the torque on the tilted, bulging Earth; the analysts of the century following, d'Alembert and Euler and their successors, ground out the period from the laws of motion, fixing the cycle's length near twenty-six thousand years and deriving the smaller nodding terms alongside it; and the nineteenth century's positional astronomers verified the whole computation against a century of star places, the discovery that began as Hipparchus's puzzle ending as the celestial mechanics' most complete demonstration, the office's succession being written down before its future holders' tenures arrived, the pole star being the one subject of this series whose future history was published before it happened.

The discovery chapter's southern closing deserves the northern reader's attention, because the two hemispheres' discoveries were asynchronous in a way the modern reader forgets. The northern wheel's hub has been watched since watching began, the circumpolar stars of the era's skies being every ancient culture's familiar furniture; the southern hub was first systematically observed by the European expeditions of the sixteenth century, and its blankness was published, argued over, and eventually accepted as a fact of the planet, the southern sky's one great discovery being an absence. The asynchrony has a practical echo still: the northern incumbent's lore is the common property of dozens of cultures across ten millennia, the southern blank's lore is the specialist property of the last five centuries' navigators and astronomers, and the volume's treatment keeps the asymmetry honest, the discovery of the pole stars being two stories told in one book, the story of a star found and the story of a star found missing.

A closing discovery note belongs to the era that made the office modern, the centuries when the pivot's motion was turned from a curiosity into an instrument. The medieval astronomers of the Islamic world carried the precession through their tables, the zij tradition keeping the pivot's place current for the calendar's needs; the Renaissance observers, Tycho's generation, remeasured the wobble with instruments an order finer than the classical era's; and the great cataloguers of the seventeenth and eighteenth centuries, Halley among them, fixed the succession's schedule on the modern footing, the office's biography becoming a working table in the ephemerides. The note completes the discovery chapter's arc, from the first watcher's wheel to the printed table of its turning, the

discovery of the pole stars having been, in its long middle chapters, the work of the world's observing institutions, the office being discovered in installments by every era that needed to know exactly where the sky's still point stood.